Bangladesh Army International University of Science and Technology

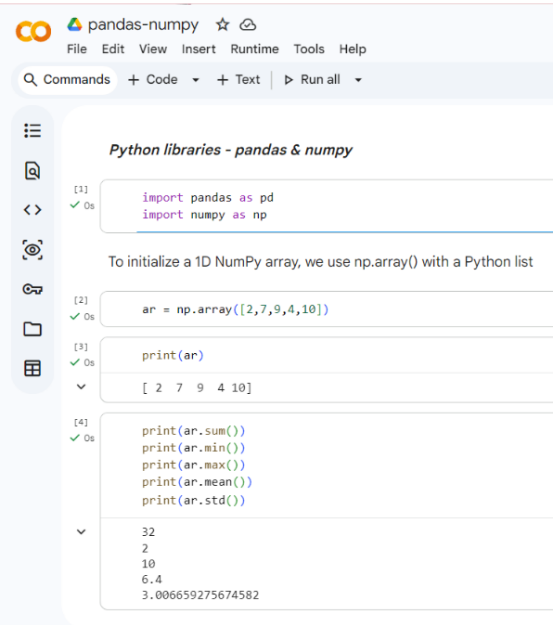
Department of Computer Science and Engineering

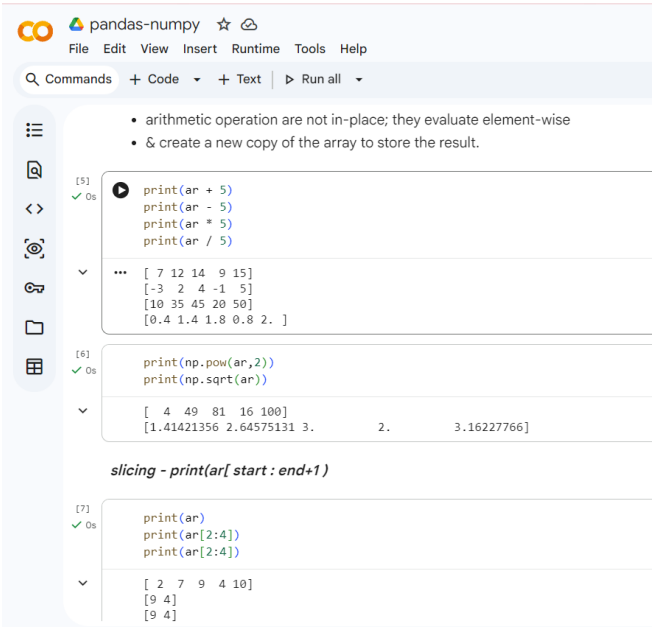
Lab Report

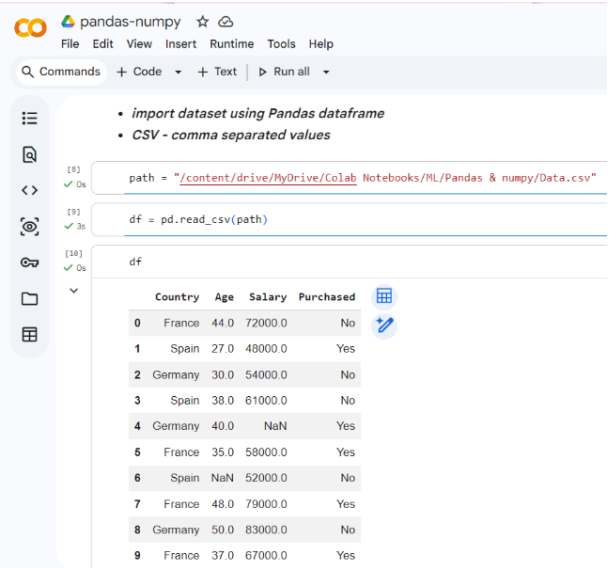
Lab Report No	:	01					
Lab Report Name	:	Preprocessing Basics					
Course Title	:	Machine Learning Sessional					
Course Code	:	CSE 412					
Name	:	Nosratee Jahan Naba					
ID	:	1118005					
Level	:	4	**Term**	: I	**Section**	: A	**Group** : 1
Date of Submission	:	11/10/2 6	**Semester**	: Fall	**Year**	: 2026	

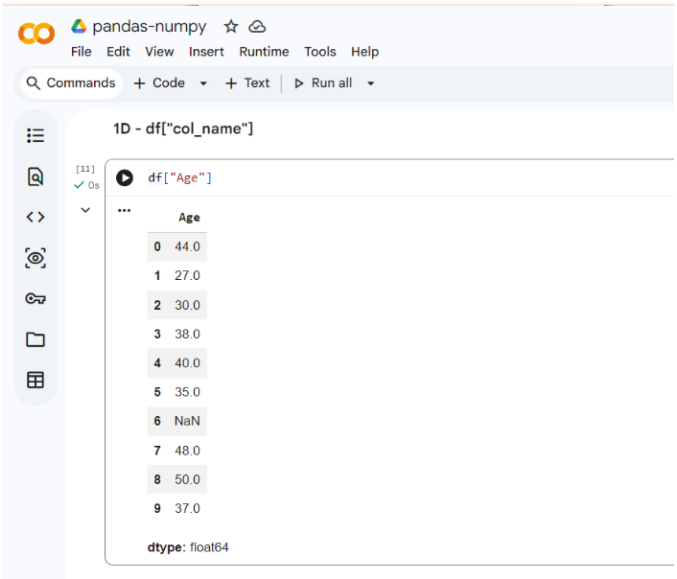
Question: Numpy, Pandas basics

Code Implementation:









pandas-numpy

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head() - gives first 5 rows by deafult

```
[13] df.head()
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes

pandas-numpy

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df.describe() - apply on numerical values,used to generate descriptive statistics

```
[16] df.describe()
```

	Age	Salary
count	9.000000	9.000000
mean	38.777778	63777.777778
std	7.693793	12265.579662
min	27.000000	48000.000000
25%	35.000000	54000.000000
50%	38.000000	61000.000000
75%	44.000000	72000.000000
max	50.000000	83000.000000

pandas-numpy

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2D - pandas DataFrame (with multiple columns)

```
[12] df[["Age", "Salary", "Country"]]
```

	Age	Salary	Country
0	44.0	72000.0	France
1	27.0	48000.0	Spain
2	30.0	54000.0	Germany
3	38.0	61000.0	Spain
4	40.0	NaN	Germany
5	35.0	58000.0	France
6	NaN	52000.0	Spain
7	48.0	79000.0	France
8	50.0	83000.0	Germany
9	37.0	67000.0	France

pandas-numpy

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df.info() - gives summary

```
[17] df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    10 non-null     object
 1   Age        9 non-null      float64
 2   Salary     9 non-null      float64
 3   Purchased  10 non-null     object
dtypes: float64(2), object(2)
memory usage: 452.0+ bytes
```

pandas-numpy

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df.head(no of rows)

```
[14] df.head(9)
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	Germany	50.0	83000.0	No

pandas-numpy

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df.isna() - is a method used to detect missing or null values in a DataFrame

- isna - is not available

```
[18] df.isna()
```

	Country	Age	Salary	Purchased
0	False	False	False	False
1	False	False	False	False
2	False	False	False	False
3	False	False	False	False
4	False	False	True	False
5	False	False	False	False
6	False	True	False	False
7	False	False	False	False
8	False	False	False	False
9	False	False	False	False

pandas-numpy

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shape - gives row & column

```
[15] df.shape

(10, 4)
```

pandas-numpy

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```
df.isna().sum()
```

	0
Country	0
Age	1
Salary	1
Purchased	0

dtype: int64

pandas-numpy

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df[df.isna().any(axis=1)] - used to filter and return all rows in a Pandas DataFrame that contain at least one missing or NaN value

[20] 0s

```
df[df.isna().any(axis=1)]
```

	Country	Age	Salary	Purchased
4	Germany	40.0	NaN	Yes
6	Spain	NaN	52000.0	No

drop age column

[21] 0s

```
df = df.drop(columns=["Age"])
```

[22] 0s

```
df.head()
```

	Country	Salary	Purchased
0	France	72000.0	No
1	Spain	48000.0	Yes
2	Germany	54000.0	No
3	Spain	61000.0	No
4	Germany	NaN	Yes